

Compiled with problems:

ERROR in ./docs/semester-1/gqur-300/unit-03/topic-01.mdx

Module build failed (from ./node_modules/@docusaurus/mdx-loader/lib/index.js):

Error: MDX compilation failed for file

["/home/a2ahs/mega_book_for_B.Ed/.claude/worktrees/agent-affcb48d7818e8183/docs/semester-1/gqur-300/unit-03/topic-01.mdx"](/home/a2ahs/mega_book_for_B.Ed/.claude/worktrees/agent-affcb48d7818e8183/docs/semester-1/gqur-300/unit-03/topic-01.mdx)

Cause: Unexpected character `=` (U+003D) before name, expected a character that can start a name, such as a letter, `\$`, or `\_`

Details:

```
{
  "column": 36,
  "file": "",
  "message": "Unexpected character `=` (U+003D) before name,
expected a character that can start a name, such as a
letter, `$`, or `_`",
  "line": 141,
  "name": "141:36",
  "place": {
    "_bufferIndex": 33,
    "_index": 0,
    "line": 141,
    "column": 36,
    "offset": 7642
  },
  "reason": "Unexpected character `=` (U+003D) before name,
expected a character that can start a name, such as a
letter, `$`, or `_`",
  "ruleId": "unexpected-character",
  "source": "micromark-extension-mdx-jsx",
  "url": "https://github.com/micromark/micromark-extension-
mdx-jsx#unexpected-character-at-expected-expect"
}
```

at compileToJSX

choosing the right operation when a daily problem does not tell you which one to use.

Explanation

Powers and roots

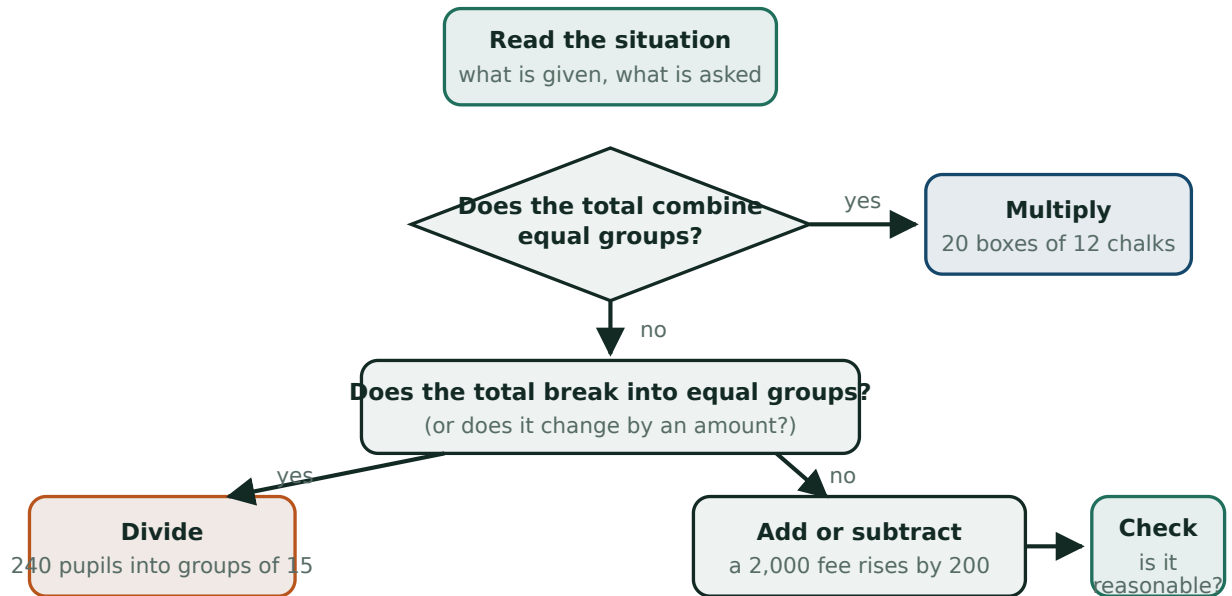
A power is repeated multiplication: 8 squared (written 8^2) means $8 \times 8 = 64$, and 2 cubed (2^3) means $2 \times 2 \times 2 = 8$. The number being multiplied is the base; the small raised number, the exponent, counts the multiplications. The square of a number is its second power, and it is genuinely a square: a square floor 8 tiles on a side holds exactly $8 \times 8 = 64$ tiles. The diagram above shows the square numbers growing as tile grids; the "squared" arrow climbs from side to area, and the square root arrow climbs back down. The square root of 64, written $\text{sqrt}(64)$, is 8: the side of the square that holds 64 tiles.

Worked example. A store room has a square floor of 50 square metres. What is the length of one side, to the nearest metre? The side is $\text{sqrt}(50)$. Since $7 \times 7 = 49$ and $8 \times 8 = 64$, $\text{sqrt}(50)$ sits between 7 and 8, much closer to 7: about 7.1 metres. Estimation first (Unit 1), exactness when needed: 7.07 metres if the carpenter must cut a skirting board.

A common misconception to correct. Trainees sometimes read 8^2 as $8 \times 2 = 16$. The exponent counts multiplications of the base, not a multiplication by itself. The tile picture is the cure: 8 squared is an 8-by-8 floor, not an 8-by-2 corridor. The same picture prevents the twin error of thinking squaring doubles: 8 squared is 64, not 16.

Applications of numbers and operations in daily life

Choosing the operation: what does the total do?



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Most daily-life arithmetic fails before any number is touched: the wrong operation is chosen. The flowchart above is a repair. Its core question is: **what does the total do?**

- If the total **combines equal groups** (20 boxes of 12 chalks each), multiply.
- If the total **breaks into equal groups** (240 pupils into groups of 15), divide.
- If the total **grows or shrinks by a change** (a fee of 2,000 rises by 200), add or subtract.

Worked example. A school orders transport for 187 pupils; each van carries 26. How many vans? The total (pupils) breaks into equal groups (vanloads), so divide: 187 divided by 26 is 7 with remainder 5. Look back: 7 vans carry 182 pupils, but 5 pupils remain, so the real-world answer is 8 vans. The look-back step is not decoration; it changes the answer.

Choosing units as part of choosing the operation. Daily problems arrive wearing units: rupees, metres, pupils per van, rupees per notebook. A rate (rupees per notebook) multiplies a count (notebooks) to give a total (rupees). Keeping the units attached through the computation - 80 rupees/notebook x 100

notebooks = 8,000 rupees - makes the operation choice almost automatic and errors visible.

Activity: Tile the courtyard

Work in groups of three. Your tutor gives each group a courtyard plan (a rectangle, say 12 metres by 9 metres) and a choice of square tile sizes (0.5 m, 1 m, 1.5 m). For each tile size, compute how many tiles fill the courtyard and how many full tiles a 64-tile budget would cover. Then decide which tile size wastes the least cutting, and defend the choice in two sentences. About 25 minutes.

Check your understanding

1. Compute: 6 squared, 9 squared, and the square roots of 81 and 100.
(Remember)
2. A square classroom floor has an area of 36 square metres. What is the length of each wall, and how do you know? (Apply)
3. Between which two whole numbers does the square root of 40 lie? Justify in one sentence. (Apply)
4. For each situation, name the operation and the reason: (a) 15 boxes of 40 books each; (b) 300 pupils into classes of 30; (c) a 5,600-rupee bill paid with 6,000 rupees. (Apply)
5. A trainee computes 187 pupils divided by 26-pupil vans as 7 and orders 7 vans. Identify the error in one sentence using the look-back idea. (Analyze)

Summary

Powers are repeated multiplication, and squares are genuinely areas: an 8-by-8 floor holds 64 tiles, which is why the square root of 64 is 8. Estimation brackets a root between neighbouring squares before any exact work. When a daily problem does not name its operation, ask what the total does: combine equal groups (multiply), break into equal groups (divide), or change by an amount (add or subtract) - then check the answer for reasonableness, because the real-world answer may need the remainder you were about to discard.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can compute squares and square roots of perfect squares, and bracket other roots between neighbouring squares.
- ☐ I can name the correct operation for a daily-life problem by asking what the total does.
- ☐ I keep units attached through a computation and use them to check the operation.

Try this at your practicum school

Find one tiled or bricked area (verandah, courtyard, walkway). Estimate its dimensions, compute its area, and work out how many square tiles of one size would cover it. Ask the caretaker how many tiles were actually used, and note any difference and its likely cause.

Summative task

The school wants a square mosaic of 225 small tiles on a corridor wall, and the wall space is a square of side 20 tiles. In a short written answer: (a) state how wide the mosaic is (the square root of 225) and whether it fits the wall, (b) the school has only these 225 tiles in stock: compute how many more tiles a full 20-by-20 design would need, and (c) propose one real reason a school might prefer the smaller mosaic, with a computation that supports it.

Mini-rubric (each criterion 0-2 marks):

- *Root and fit*: the mosaic is 15 by 15 and fits inside 20 by 20, with reasoning shown.
- *Tile count*: a 20-by-20 design uses 400 tiles, so 175 more tiles would be needed; the comparison with 225 is computed correctly.
- *Supported proposal*: the reason is argued with a computation (for example, cost per tile or leftover tiles for repairs), not merely asserted. **(Analyze)**

Further reading

- Marecek, L., Anthony-Smith, M., & Mathis, M. H. (2020). *Prealgebra 2e* (Ch. 5, Section 5.7 Simplify and Use Square Roots; Ch. 9 Math Models and Geometry). OpenStax. <https://openstax.org/details/books/prealgebra-2e>
- Gula, T., & Lovric, M. (2025). Promoting mathematical thinking of university students: The case of a numeracy course. *Canadian Journal of Science, Mathematics and Technology Education*. <https://eric.ed.gov/?id=EJ1489427>

Tip: select any text on this page first to comment on that specific passage - or leave nothing selected for general feedback.

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